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BSIT 2E-G2

Activity 8: Database Backup and Recovery

Learning Objectives

After completing this activity, students should be able to:

1. Identify different types of **database backups**
2. Explain how **backup and recovery protect data**
3. Perform **basic backup and restore operations** in SQL Server

PART 1 — DISCUSSION

What is Database Backup?

A **backup** is a copy of the database stored in another location. Purpose:

- Prevent data loss
- Recover from system failure
- Restore accidentally deleted data

Why is Backup Important?

Example (Library System):

If the database crashes:

All books, members, and borrow records are lost

With backup:

You can restore everything

Types of Backup (SQL Server)

1. Full Backup

- Copies the **entire database**

Example:

```
BACKUP DATABASE FinalLibraryDB
TO DISK = 'C:\Backup\Library_Full.bak';
```

Easy to restore
Takes more storage
2. Differential Backup

- Copies only changes since last full backup

Faster
Smaller size

3. Transaction Log Backup

- Records all transactions

Point-in-time recovery
Advanced usage

Recovery Models

Model	**Description**
Simple	No log backup SQL Server does NOT keep transaction history for recovery.
Full	Complete recovery SQL Server records every change until you back it up.

For beginners:

Use **FULL recovery model**

PART 2 — UNDERSTANDING

1. What is the purpose of a database backup?
   — Database backup serves as a copy of a database, used to recover data in case of database deletion or corruption of data.
2. Which backup type copies the entire database?
   — The backup type that copies the entire database is the full backup.

3. Which backup is faster: Full or Differential?
   - Differential backup is the faster one, since it only backup the changes made since the last full backup instead of copying the entire database again.
4. Why is recovery important?
   - Recovery is important because it has the ability to bring back lost or corrupted data that is important for the system.

PART 3 — HANDS-ON ACTIVITY

--Step 1 Set Recovery Model - This allows advanced recovery like restoring specific

points in time.

```
ALTER DATABASE FinalLibraryDB
SET RECOVERY FULL;
```

Use the query below if you encounter SINGLE USER ISSUE

ALTER DATABASE FinalLibraryDB SET MULTI_USER;

--Step 2 Perform FULL BACKUP - This creates a full copy of your database.

```
BACKUP DATABASE FinalLibraryDB
TO DISK = 'C:\Backup\FinalLibraryDB_Full.bak'
WITH INIT;
```

Step 3 — Verify Backup

Check folder:

C:\Backup\

You should see: FinalLibraryDB_Full.bak

Step 4 — Simulate Data Loss

Run:

```
DELETE FROM Books;
```

What just happened?

- All the data/records from the books table is deleted.

Step 5 — Restore Database - We are restoring the database from backup.

```
USE master;
```

```
RESTORE DATABASE FinalLibraryDB
FROM DISK = 'C:\Backup\FinalLibraryDB_Full.bak'
WITH REPLACE;
```

If you encounter error like lock and you cannot connect… close the database that you need to restore.

Step 6 — Verify Recovery

```
USE FinalLibraryDB;

SELECT * FROM Books;
```

Data is back!

PART 4 — ANALYSIS QUESTIONS

1. What happened after deleting the Books table data?
   - All the data/records from the Books table is deleted and gone.
2. How did backup help recover the data?
   - The backup is copy of the last saved database, and is used to restore lost data/records.
3. What would happen if no backup existed?
   - If the backup doesn't exist, the deleted data/records cannot be recovered again and be gone forever. We can't afford to lose these data in a working system, which is why backup is important.

PART 5 — MINI PROJECT APPLICATION

Scenario:

Your **Library System** is used daily.

Problem: A librarian accidentally deletes all BorrowedBooks

Task:

Students must:

1. Create a backup
2. Simulate deletion
3. Restore the database

Write your SQL:

Backup:

BACKUP DATABASE LibraryDB

TO DISK = 'C:\Users\rronf\OneDrive\Desktop\Backup\LibraryDB_Full.bak'

WITH INIT;

Delete:

DELETE FROM BorrowedBooks;

Restore:

USE master;

RESTORE DATABASE LibraryDB

FROM DISK = 'C:\Users\chunmr\OneDrive\Desktop\Backup\LibraryDB_Full.bak'

WITH REPLACE;

PART 6 — FILL IN THE BLANKS

1. A **Backup** is a copy of the database used for recovery.
2. A **Full** backup copies the entire database.
3. A **Differential** backup copies only changes since last backup.
4. The **Full Recovery** model allows full recovery.
5. The command used to create backup is **BACKUP DATABASE [DATABASE NAME]**.
6. The command used to restore database is **RESTORE DATABASE [DATABASE NAME]**.

```
CREATE DATABASE DemoLibraryDB;
USE DemoLibraryDB;

CREATE TABLE Books (
 BookID INT PRIMARY KEY,
 Title VARCHAR(50),
 Quantity INT
);

INSERT INTO Books VALUES
(1,'Database Systems',5),
```

```
(2,'Networking Basics',3),
(3,'AI Fundamentals',4);

SELECT * FROM Books;

BACKUP DATABASE DemoLibraryDB
TO DISK = 'C:\SQLBackup\DemoLibraryDB.bak'
WITH INIT;

USE master;
RESTORE DATABASE DemoLibraryDB
FROM DISK = 'C:\SQLBackup\DemoLibraryDB.bak'
WITH REPLACE;
```